

Topic: DOM Events
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What is DOM

How can we perform DOM Manipulation

1. Access the elements
2. Change style and attributes
3. Change content - HTML or the text content
4. Add, remove or reorder the elements
5. Listen to and react to user events

```
document.getElementsByTagName('tag')  
document.getElementsByClassName('className')  
document.getElementById('id')
```

```
let divList = document.getElementsByTagName('div')  
// assume we have 5 div elements  
divList[2].style.backgroundColor = 'blue'
```

```
set attributes for an element using setAttribute('attributeName', 'attributeValue')  
get attribute for an element using getAttribute('attributeName')
```

innerHTML and innerText

- using innerHTML we can add complex HTML code
- using innerText we can pass in the exact value that we have to display

classList - which stores the list of all the classes applied to the element
we can add, remove or toggle the classes

classList.add('className') - adding the class to the list

classList.remove('className') - removing the class from the list

classList.toggle('className') - to remove if present and to add if missing

classList.contains('className') - to check whether the class is applied or not

4. Add, remove or reorder the elements

a. Adding Elements

- createElement
- append
- appendChild
- insertBefore
- innerHTML

	append
. accept Node and strings	
. it accepts multiple args	
. returns undefined	

appendChild
accepts only Node
accepts a single argument
returns a node

createElement - creates a new element using document reference
append - can add multiple nodes / strings to the structure
appendChild - can add single node to the structure
innerHTML - can update the complete HTML structure for referenced node
insertBefore - it will insert the new element before the referenced element

b. Remove Elements

- remove
- removeChild
- innerHTML

innerHTML - can update the complete HTML structure
remove - it removes the element from the DOM.
removeChild - it removes the referenced child from the referenced parent node

c. Reorder Elements

to reorder the elements use insertBefore, innerHTML, append or appendChild.

5. Events in DOM

whatever user is performing on the webpage is an event

- scroll
- mouse movement
- clicks
- key press

- text inputs

example@example - invalid email address

using fake email ids - other than gmail, outlook, hotmail, yahoo, yandex

prevent user from clicking right click to perform any illicit operation

Listen - there is an event listener which is waiting for an input from the user
- once the user interacts, it is made available to the JS

React - based on the user interaction we execute a set of instructions
- the response is relayed to the user

TO add an event listener to a Node

- first identify the node
- add the event listener using `addEventListener()`
- pass the event that is to be captured
 - click
 - key press
 - scroll
 - key release
- perform the set of instructions

```
addEventListener('event', function(){ //steps to execute })
```

1. Mouse Events

- click
- dblclick
- mouseenter / mouseleave
- mousedown / mouseup
- mousemove

2. Keyboard Events

- keydown
- keyup
- keypress - deprecated

3. Form Events

- submit
- input
- change
- focus / blur